

Project 2: Jacobians

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Prompt: The key property that allows us to define a group structure on a cubic curve is Bezout's theorem that indicates that there are precisely three intersection points of a cubic curve with a line. The goal of this project is to learn a little bit about higher-dimensional abelian varieties which give a way to associated a group structure with points on other sorts of curves. Research and write a report that addresses the following questions: (a) What are divisors and principal divisors on a curve, (b) What is the Jacobian of a curve and how is its group structure defined? (c) How do you explicitly compute with the Jacobian of a genus 2 curve? (d) How is an elliptic curve a Jacobian? and (e) What can be said about the Mordell-Weil group of a Jacobian?

§1 Introduction

Note: I will attempt to remain rigorous while suppressing mention of “varieties” and the fields over which they may be defined in favor of the language of “curves” in an attempt to demonstrate understanding of the material without hiding behind concepts I don't fully understand the significance of.

§2 Divisors and principal divisors

Let C be a projective curve. Then a *divisor* of C is a finite formal sum

$$\sum_{i=1}^m n_i P_i, \tag{1}$$

where each P_i is a point on C and each $n_i \in \mathbb{Z}$. In other words, the set of divisors is the free abelian group generated by the points of C . This group is called the *divisor group* of C (denoted $\text{Div}(C)$), and addition is defined as usual for linear combinations, where the coefficients of “like terms” (i.e. terms corresponding to the same point) are added in the final sum.

We now need the concept of the order of a function, whose definition seems to get further into the weeds of valuation rings than the rest of this project will require. As such, I will include a couple of theorems about the order $\text{ord}_P(f)$ of a function f at a point P which establish the properties we will need:

1. If $\text{ord}_P(f) > 0$, then f has a zero at P . If $\text{ord}_P(f) < 0$, then f has a pole (or, is infinite) at P . If $\text{ord}_P(f) = 0$, then f is defined but nonzero at P .
2. Given a smooth curve C and a rational function f defined over C , there are only finitely many points of C at which f has a pole or a zero. Thus $\text{ord}_P(f) = 0$ at all but finitely many points $P \in C$.

Given that C is smooth, this allows us to assign to any rational function f over C a divisor $\text{div}(f)$ defined by

$$\text{div}(f) = \sum_{P \in C} \text{ord}_P(f) \cdot P.$$

We can now define a *principal divisor* to be any divisor $D \in \text{Div}(C)$ such that $D = \text{div}(f)$ for some rational function f defined over C .

§3 The Jacobian of a curve

Two divisors $D_1, D_2 \in \text{Div}(C)$ are said to be *linearly equivalent*, denoted $D_1 \sim D_2$, if $D_1 - D_2$ is principal. We define the Picard group $\text{Pic}(C)$ to be the quotient of $\text{Div}(C)$ by the subgroup of principal divisors, i.e. divisors modulo linear equivalence. The *degree* of a divisor is the sum of its coefficients:

$$\deg \left(\sum_{i=1}^m n_i P_i \right) = \sum_{i=1}^m n_i. \quad (2)$$

We can then define the groups $\text{Div}^0(C)$ and $\text{Pic}^0(C)$ to be the subgroups of $\text{Div}(C)$ and $\text{Pic}(C)$ respectively consisting of the elements of degree zero. The Jacobian $J(C)$ is then defined to be $\text{Pic}^0(C)$, or in other words, the degree 0 divisors of C modulo linear equivalence. In fact, it seems to often be defined more abstractly in terms of certain functors, but then the construction is proven to be isomorphic to $\text{Pic}^0(C)$, so we have opted to use that as the definition here. The group structure is thus the usual group structure in a quotient group:

$$[D_1] + [D_2] = [D_1 + D_2].$$

§4 Jacobians of genus 2 curves

§5 Elliptic curves are Jacobians

§6 The Mordell-Weil group of a Jacobian

§7 Bibliography

The Arithmetic of Elliptic Curves by Joseph H. Silverman
Used for understanding the divisor group of a curve.

Algebraic Geometry by Robin Hartshorne
Used for understanding the formulation of the Jacobian in terms of $\text{Pic}^0(C)$.